

Microlocal Morse Theory and Truncated Microsupport

Toshi2019

December 2, 2025*

Preface

In these notes, We study microlocal Morse theory, which is developed by Guillermou, Kashiwara, and Schapira in [GKS12], but replace microsupports with truncated microsupprts introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03].

1 Truncated Microsupport

Definition 1.1. (i) Let $F \in D^b(\mathbf{k}_X)$. The closed conic subset $SS_k(F)$ of T^*X is defined by: $p \notin SS_k(F)$ if and only if F satisfies the equivalent conditions in Prop.

(ii) aa

2 Results

Theorem 2.1. Let F in $D^b(\mathbf{k}_M)$, and $\psi: M \rightarrow \mathbf{R}$ be a real-valued function of class C^1 , proper on $\text{supp}(F)$. Let $a < b$ be real numbers. We assume $d\psi(x) \notin SS_k(F)$ for $a \leq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}(]-\infty, t[)$. Then, the restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each $j \leq k$.

Lemma 2.2. Let $-\infty < a < b < +\infty$ be real numbers and $G \in D^b(\mathbf{R})$. Let k be an integer. If

$$\left(H^j_{[t, +\infty[}(G) \right)_t \cong 0 \quad \text{for all } t \in [a, b[, \text{ and all } j \leq k.$$

Then one has

$$H^j(]-\infty, b[; G) \xrightarrow{\sim} H^j(]-\infty, a[; G)$$

for all $j < k$.¹

*2025/03/09 first draft

¹For k , this morphism might be injective. (not written in this draft.)

Proof. Let $t \in [a, b[$. We will denote by I_t the open interval $] -\infty, t[$ and denote by Z_t the closed interval $] -\infty, t] (= \overline{I_t})$. By applying to the distinguished triangle

$$\mathrm{R}\Gamma_{\mathbf{R}-I_t}(G) \rightarrow G \rightarrow \mathrm{R}\Gamma_{I_t}(G) \xrightarrow{+1}$$

the functor $(\cdot)_{\overline{I_t}}$, we have

$$(\mathrm{R}\Gamma_{[t, +\infty[}(G))_{\{t\}} \rightarrow G_{Z_t} \rightarrow \mathrm{R}\Gamma_{I_t}(G) \xrightarrow{+1}.$$

Applying the functor $\mathrm{R}\Gamma(\mathbf{R}; \cdot)$ to this, we have

$$(2.1) \quad (\mathrm{R}\Gamma_{[t, +\infty[}(G))_t \rightarrow \mathrm{R}\Gamma(Z_t; G) \rightarrow \mathrm{R}\Gamma(I_t; G) \xrightarrow{+1}.$$

Then consider the following conditions:

$$\begin{aligned} (a)_s^j & \quad \varinjlim_{t>s} H^j(I_t; G) \xrightarrow{\sim} H^j(I_s; G) \quad \text{for all } s \in [a, b[, \\ (b)_t^j & \quad \varprojlim_{s<t} H^j(I_t; G) \xleftarrow{\sim} H^j(I_s; G) \quad \text{for all } t \in]a, b]. \end{aligned}$$

By hypothesis, it follows from (2.1) that $H^j(Z_t; G) \xrightarrow{\sim} H^j(I_t; G)$ for $t \in [a, b[$. Similarly, we have injections $H^k(Z_t; G) \hookrightarrow H^k(I_t; G)$ for $t \in [a, b[$. Since $\varinjlim_{t>s} H^j(I_t; G) \cong H^j(Z_s; G)$, $(a)_s^j$ holds for any $j < k$. Similarly,

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(Z_s; G)$$

gives injections

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(I_s; G)$$

for all $s \in [a, b[$. Moreover, $(b)_t^j$

□

Proof of Theorem 2.1. First, we have

$$H^j(M_t; F) \cong H^j \mathrm{R}\Gamma(] -\infty, t[; \mathrm{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leq \psi(x) < b$, and by the formula $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} \mathrm{SS}_k(F)$, we have

$$(H^j \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F))_{\psi(x)} = 0$$

for each $j \leq k$. Therefore, letting $d \in \mathbf{Z}$ be the minimum number that

$$H^d \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \neq 0,$$

by the long exact sequence

$$\begin{aligned} 0 & \rightarrow H^d \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow H^d \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \rightarrow H^d \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \\ & \rightarrow H^{d+1} \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow H^{d+1} \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \rightarrow H^{d+1} \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \\ & \rightarrow \cdots \\ & \rightarrow H^k \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow H^k \mathrm{R}\Gamma_{]-\infty, b[}(\mathrm{R}\psi_* F) \rightarrow H^k \mathrm{R}\Gamma_{]-\infty, a[}(\mathrm{R}\psi_* F) \\ & \rightarrow H^{k+1} \mathrm{R}\Gamma_{[a, b[}(\mathrm{R}\psi_* F) \rightarrow \cdots, \end{aligned}$$

we have isomorphisms

$$(H^j \mathbf{R}\Gamma_{]-\infty, b[}(\mathbf{R}\psi_* F))_{\psi(x)} \xrightarrow{\sim} (H^j \mathbf{R}\Gamma_{]-\infty, a[}(\mathbf{R}\psi_* F))_{\psi(x)}$$

for any $j \leq k$ and $a \leq \psi(x) < b$, and then,

$$(2.2) \quad H^j \mathbf{R}\Gamma_{]-\infty, b[}(\mathbf{R}\psi_* F) \xrightarrow{\sim} H^j \mathbf{R}\Gamma_{]-\infty, a[}(\mathbf{R}\psi_* F)$$

for all $j \leq k$. Then we can conclude that

$$H^j(M_b; F) \xrightarrow{\sim} H^j(M_a; F)$$

by applying to the equality (2.2) the functor $\Gamma(]-\infty, b[; \cdot)$ and calculating as follows:

$$\begin{aligned} \Gamma(]-\infty, b[; H^j \mathbf{R}\Gamma_{]-\infty, b[}(\mathbf{R}\psi_* F)) &\cong \Gamma_{]-\infty, b[}(]-\infty, b[; \mathbf{R}^j \psi_* F) \\ &\cong \Gamma(]-\infty, b[; \mathbf{R}^j \psi_* F) \\ &\cong H^j(\psi^{-1}(]-\infty, b[); F) \\ &\cong H^j(M_b; F) \end{aligned}$$

and

$$\begin{aligned} \Gamma(]-\infty, b[; H^j \mathbf{R}\Gamma_{]-\infty, a[}(\mathbf{R}\psi_* F)) &\cong \Gamma_{]-\infty, a[}(]-\infty, b[; \mathbf{R}^j \psi_* F) \\ &\cong \Gamma(]-\infty, a[; \mathbf{R}^j \psi_* F) \\ &\cong H^j(\psi^{-1}(]-\infty, a[); F) \\ &\cong H^j(M_a; F). \end{aligned}$$

□

References

- [GKS12] Stéphane Guillermou, Masaki Kashiwara, Pierre Schapira, *Sheaf Quantization of Hamiltonian Isotopies and Applications to Nondisplaceability Problems*, Duke. Math. J. 161, No. 2, pp.201–245, 2012.
- [KMS03] Masaki Kashiwara, Teresa Monteiro Fernandes, Pierre Schapira, *Truncated Microsupport and Holomorphic Solutions of D-modules*, Ann. Scient. Éc. Norm. Sup., série 4, tome 36, pp.583–399, 2003.
- [KS90] Masaki Kashiwara, Pierre Schapira, *Sheaves on Manifolds*, Grundlehren der Mathematischen Wissenschaften, 292, Springer, 1990.
- [ST92] Pierre Schapira, Nobuyuki Tose, *Morse Inequalities for $\mathbf{R}$ -constructible Sheaves*, Adv. Math. 93, pp.1–8, 1992.